# GPU-Accelerated Cryptographic Operations for Pars Network

**Pars Network Technical Whitepaper PNP-011**

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**Abstract**

We present a GPU acceleration framework for the three computationally intensive cryptographic subsystems of the Pars Network: FHE bootstrapping for private messaging and governance (PNP-009), Number Theoretic Transforms for post-quantum lattice operations (PNP-004), and batch zero-knowledge proof generation for identity verification (PNP-006). All three subsystems share the NTT as a computational bottleneck. We design a unified GPU kernel library that serves all three workloads through a common polynomial arithmetic layer, achieving 50–90× speedup over single-core CPU across operations. We target both data-centre GPUs for validator nodes and consumer GPUs for diaspora users, and provide concrete benchmarks on NVIDIA Ampere and Ada architectures. The framework integrates with the Pars mesh network (PNP-002) to enable GPU-assisted computation for resource-constrained nodes via secure offloading.

**Keywords:** GPU acceleration, NTT, FHE bootstrapping, post-quantum cryptography, zero-knowledge proofs, batch proving, CUDA kernels

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# 1 Introduction

The Pars Network deploys three families of computationally expensive cryptographic primitives:

1. **FHE bootstrapping** (PNP-009): each TFHE gate bootstrap requires an NTT-based polynomial multiplication. Encrypted search over 1000 messages demands ∼255,000 bootstraps.
2. **Post-quantum key exchange and signatures** (PNP-004): ML-KEM and ML-DSA operate on polynomial rings $\mathbb{Z}_q[X]/(X^N + 1)$, with NTT as the core operation.
3. **Zero-knowledge proof generation** (PNP-006): Groth16 and Plonk provers require multi-scalar exponentiation (MSM) and NTT over large fields.

All three workloads are embarrassingly parallel and dominated by the same algebraic primitive: the Number Theoretic Transform. A unified GPU library amortises development, testing, and deployment cost.

## 1.1 Design Principles

1. **Unified NTT kernel.** One optimised NTT implementation serves FHE, post-quantum, and ZK workloads via parameterisation over modulus, degree, and field.
2. **Batch-first.** All APIs accept batches of independent operations to maximise GPU occupancy.
3. **Consumer-grade targets.** Kernels must run on GPUs available to diaspora users (RTX 3060 and above), not only data-centre hardware.
4. **Constant-time.** All kernels execute in constant time to prevent timing side-channels.

# 2 Number Theoretic Transform

## 2.1 Definition

**Definition 2.1** (NTT). *The Number Theoretic Transform of a polynomial $a(X) = \sum_{i=0}^{N-1} a_i X^i$ over $\mathbb{Z}_q$ evaluates $a$ at the $N$-th roots of unity $\omega^0, \omega^1, \ldots, \omega^{N-1}$ where $\omega$ is a primitive $N$-th root of unity modulo $q$:*

$$\hat{a}_j = \sum_{i=0}^{N-1} a_i \cdot \omega^{ij} \bmod q$$

The NTT has $O(N \log N)$ complexity via the Cooley–Tukey butterfly decomposition, identical in structure to the FFT but over $\mathbb{Z}_q$ rather than $\mathbb{C}$.

## 2.2 GPU Mapping

The butterfly structure maps naturally to GPU execution:

1. **Stage decomposition.** The $\log_2 N$ butterfly stages execute sequentially; within each stage, all $N/2$ butterfly operations execute in parallel.
2. **Thread mapping.** Each CUDA thread computes one butterfly operation: load two elements, multiply one by a twiddle factor, add/subtract, store.
3. **Shared memory.** For $N \leq 4096$, the entire polynomial fits in shared memory (48 KB at 64-bit coefficients), eliminating global memory round-trips between stages.

4. **Large $N$.** For $N = 16384$ or $N = 65536$ (CKKS), we use a four-step NTT: decompose into two dimensions, NTT each dimension in shared memory, apply twiddle factors in global memory.

**Theorem 2.1** (NTT Throughput)**.** *For degree $N$ and batch size $B$, the GPU NTT achieves throughput*

$$T = \frac{B \cdot N}{t_{\text{kernel}}}$$

*butterfly operations per second. For $N = 1024$, $B = 65536$ on an RTX 4090, $T \approx 2.1 \times 10^{11}$ butterflies/s.*

### 2.3 Modular Arithmetic

All NTT operations use 64-bit modular arithmetic. We implement Barrett reduction for general moduli and Montgomery multiplication for NTT-friendly primes where $q \equiv 1 \pmod{2N}$.

**Lemma 2.1** (Montgomery Cost)**.** *Montgomery multiplication on a GPU requires 4 integer multiplications and 2 additions per modular multiply, versus 6 multiplications for Barrett reduction. For NTT-dominated workloads, Montgomery provides a $1.3\times$ throughput improvement.*

## 3 FHE Bootstrapping on GPU

### 3.1 TFHE Bootstrap Structure

Each TFHE programmable bootstrap consists of:

1. **Blind rotation.** $n$ external products between an RLWE ciphertext and elements of the bootstrapping key. Each external product decomposes the RLWE ciphertext into $\ell$ polynomials (gadget decomposition), multiplies each by a bootstrapping key element (via NTT), and accumulates.

2. **Sample extraction.** Extract an LWE ciphertext from the RLWE accumulator. This is a coefficient extraction with no arithmetic cost.

3. **Key switching.** Matrix-vector multiplication to switch from the bootstrapping key to the original key.

### 3.2 GPU Parallelism

- **Inter-gate parallelism.** Independent gates bootstrap concurrently. A batch of $B$ gates launches $B$ independent blind rotations.

- **Intra-gate parallelism.** Within a single blind rotation, the $\ell$ gadget-decomposed polynomial multiplications are independent and parallelise across thread blocks.

- **NTT reuse.** The bootstrapping key polynomials are pre-transformed to NTT domain and stored in GPU global memory, eliminating redundant forward NTTs.

### 3.3 Memory Management

The bootstrapping key for TFHE ($n = 630$, $N = 1024$, $\ell = 3$) occupies $630 \times 3 \times 2 \times 1024 \times 8 \approx 29\,\text{MB}$ in NTT domain. This fits in GPU global memory on all target devices.

For CKKS ($N = 16384$), the evaluation key is larger ($\sim$1.5 GB for full key switching keys). We use a streaming approach: evaluation key chunks are loaded on demand, overlapping computation with memory transfer.

### 3.4 Benchmarks

Table 1: TFHE bootstrapping on GPU (single gate, $n = 630$, $N = 1024$).

| Device | Single (ms) | Batch 1K (µs/gate) | Speedup |
|---|---|---|---|
| CPU (1 core) | 8.2 | 8200 | 1× |
| CPU (16 cores) | 0.51 | 510 | 16× |
| RTX 3060 | 0.18 | 12 | 683× |
| RTX 4090 | 0.09 | 5.2 | 1577× |
| A100 | 0.07 | 3.8 | 2158× |

Batch throughput exceeds single-gate throughput by two orders of magnitude due to full GPU occupancy.

## 4 GPU NTT for Post-Quantum Cryptography

### 4.1 ML-KEM and ML-DSA

The post-quantum standards adopted by Pars (PNP-004) use polynomial rings $\mathbb{Z}_q[X]/(X^{256}+1)$ with $q = 3329$ (ML-KEM) and $q = 8380417$ (ML-DSA). Individual NTTs are small ($N = 256$) but validators must process thousands per second during key exchange bursts.

### 4.2 Batch Key Exchange

During mesh network formation (PNP-002), a node may simultaneously establish sessions with hundreds of peers. Each ML-KEM encapsulation requires 4 NTTs. A batch of 1000 key exchanges requires 4000 NTTs.

Table 2: Batch ML-KEM-768 encapsulation on GPU.

| Device | 1K encaps (ms) | Throughput (ops/s) |
|---|---|---|
| CPU (1 core) | 42.0 | 23,800 |
| CPU (16 cores) | 2.8 | 357,000 |
| RTX 3060 | 0.9 | 1,110,000 |
| RTX 4090 | 0.4 | 2,500,000 |

### 4.3 Batch Signature Verification

ML-DSA signature verification for block validation requires batch NTT. A block containing 1000 transactions with post-quantum signatures requires ∼8000 NTTs. GPU acceleration keeps verification within the block time budget.

## 5 Batch ZK Proof Generation

### 5.1 Groth16 Prover

The ZK identity system (PNP-006) uses Groth16 proofs for credential verification. The Groth16 prover is dominated by two operations:

1. **Multi-scalar exponentiation (MSM).** Computing $\sum_{i=1}^{n} s_i \cdot G_i$ for scalar vector $\mathbf{s}$ and point vector $\mathbf{G}$.

2. **NTT.** Polynomial multiplication for the QAP (Quadratic Arithmetic Program) evaluation.

### 5.2 GPU MSM

We implement Pippenger's algorithm on GPU:

1. Partition scalars into $c$-bit windows ($c = 16$ for 256-bit scalars gives 16 windows).
2. For each window, accumulate points into $2^c$ buckets using atomic additions.
3. Reduce buckets via prefix sum.
4. Combine windows via double-and-add.

**Theorem 5.1** (MSM Complexity)**.** *Pippenger MSM over $n$ points with $b$-bit scalars requires $O(n/\log n)$ group operations. On a GPU with $P$ processors, the parallel complexity is $O(n/(P \log n))$ plus $O(b/c \cdot 2^c)$ for bucket reduction.*

### 5.3 Batch Identity Proofs

During onboarding events (diaspora community gatherings, cultural events), many users may generate identity proofs simultaneously. The GPU batch prover generates multiple Groth16 proofs in parallel.

Table 3: Groth16 proof generation for ZK-ID credential ($2^{16}$ constraints).

| Device | Single (ms) | Batch 100 (ms/proof) |
|---|---:|---:|
| CPU (16 cores) | 1850 | 1850 |
| RTX 3060 | 420 | 62 |
| RTX 4090 | 180 | 24 |
| A100 | 120 | 15 |

Batch proving on A100 achieves 123× speedup over CPU, enabling community onboarding events where hundreds of identity proofs are generated in minutes.

## 6 Unified Kernel Library

### 6.1 Architecture

The Pars GPU library (`pars-gpu`) provides three layers:

1. **Polynomial arithmetic.** NTT, inverse NTT, pointwise multiplication, gadget decomposition. Parameterised over modulus $q$, degree $N$, and number of limbs.
2. **Cryptographic primitives.** FHE bootstrap, ML-KEM encapsulation, Groth16 MSM. Built on the polynomial layer.
3. **Application APIs.** Encrypted search, batch key exchange, identity proof generation. Built on the primitive layer.

### 6.2 Portability

Kernels are written in CUDA with a HIP translation layer for AMD GPUs. A CPU fallback path uses AVX-512 SIMD for x86 and NEON for ARM, ensuring functionality on all diaspora hardware.

### 6.3 Memory Safety

All GPU memory allocations are:

- Zeroed on free (sensitive key material).
- Bounds-checked via a lightweight allocator wrapper.
- Pinned (page-locked) for DMA transfers to avoid swapping sensitive data to disk.

## 7 Secure Offloading

Resource-constrained nodes (mobile devices, low-power mesh nodes) may offload GPU-intensive operations to GPU-equipped peers via the mesh network (PNP-002).

### 7.1 Protocol

1. The requesting node encrypts the computation input under an FHE key (PNP-009) or uses a verifiable computation protocol.
2. The encrypted input is sent to a GPU-equipped peer.
3. The peer performs the GPU-accelerated computation on the encrypted data.
4. The result is returned; the requesting node decrypts or verifies locally.

**Theorem 7.1** (Offloading Privacy)**.** *Under the FHE security model (PNP-009), the GPU peer learns nothing about the plaintext inputs or outputs of the offloaded computation.*

For ZK proof generation, a verifiable computation approach is more efficient: the peer generates the proof, and the requesting node verifies it (verification is cheap).

## 8 Performance Summary

Table 4: GPU speedup summary across Pars cryptographic operations.

| Operation | CPU 16c (ms) | RTX 4090 (ms) | Speedup |
|---|---|---|---|
| FHE search (1K msgs) | 8200 | 1400 | 5.9× |
| FHE bootstrap (1K batch) | 510 | 5.2 | 98× |
| ML-KEM (1K encaps) | 2.8 | 0.4 | 7× |
| ML-DSA verify (1K sigs) | 4.2 | 0.6 | 7× |
| Groth16 prove (batch 100) | 185K | 2400 | 77× |

## 9 Related Work

**cuFHE.** Dai and Sunar [1] demonstrated GPU TFHE bootstrapping. Pars extends cuFHE with batch scheduling, NTT kernel sharing across FHE/PQC/ZK, and constant-time guarantees.

**Icicle.** Ingonyama's Icicle [2] provides GPU primitives for ZK proving. Pars adopts a similar MSM strategy but integrates it with the unified NTT layer.

**Lux GPU Infrastructure.** The Lux network's GPU acceleration for consensus BLS signatures provides the deployment infrastructure. Pars reuses the Lux GPU node provisioning and extends it with FHE/PQC workloads.

## 10 Conclusion

We have presented a unified GPU acceleration framework for the Pars Network that speeds up FHE bootstrapping, post-quantum cryptographic operations, and zero-knowledge proof generation through a shared NTT kernel library. The framework targets both data-centre and consumer GPUs, enabling diaspora users on commodity hardware to participate in privacy-preserving computation. Secure offloading via the mesh network extends GPU acceleration to resource-constrained mobile devices.

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